

Assignment

November 21, 2025

```
[1]: %reload_ext autoreload
      %autoreload 2
```

```
[2]: import torch
      import torch.nn as nn
      from torch.nn import functional as F
      import math
      from torch.nn import LayerNorm
```

```
[3]: from kret_studies import *
      from kret_studies.notebook import *
      from kret_studies.complex import *
```

Loaded environment variables from
/Users/Akseldkw/Desktop/Columbia/COMS4776/.env.

INFO:datasets:JAX version 0.7.2 available.

```
[4]: from kret_studies.kret_torch.mixin.constants import DEVICE_TORCH_STR,
      ↪DEVICE_LITERAL
```

```
device: DEVICE_LITERAL = DEVICE_TORCH_STR
NUM_EPOCHS = 25
```

```
[5]: from trainer import save_model_auto, load_model_auto, model_filename
```

```
[6]: # !pip install thop
      # !pip install tqdm
```

1 Part 1: Attention and Encoder-Decoder Models

In this section, you'll implement two forms of attention and use them in an encoder-decoder style transformer. You'll then compare performance based on changing dataset size and hidden dimension.

1.1 Step 1: Scaled Dot Product Attention

Implement both the base `scaled_dot_attention` method as well as the `forward` method for the Attention module.

```
[7]: def scaled_dot_attention(
    q: torch.Tensor, k: torch.Tensor, v: torch.Tensor, mask: torch.Tensor | None
    ↪ None = None
):
    """Computes scaled dot product attention with an optional mask.
    q : shape ( batch, k, hidden_size)
    k : shape ( batch, seq_len, hidden_size)
    v : shape ( batch, seq_len, hidden_size)
    mask: optional. shape (k, seq_len)

    returns:
        context : shape (batch, k, hidden_size)
        attention : shape (batch, k, seq_len)
    """

    # -----
    # FILL THIS IN
    # -----
    unnorm_attn = torch.matmul(q, k.transpose(-2, -1)) / math.sqrt(q.size(-1))
    if mask is not None:
        unnorm_attn = unnorm_attn.masked_fill(mask == 0, float("-inf"))
    attention = F.softmax(unnorm_attn, dim=-1)
    context = torch.matmul(attention, v)
    return context, attention
```

```
[8]: class Attention(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.Q = nn.Linear(hidden_size, hidden_size, bias=False)
        self.K = nn.Linear(hidden_size, hidden_size, bias=False)
        self.V = nn.Linear(hidden_size, hidden_size, bias=False)
        self.hidden_size = hidden_size
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor, annots: torch.Tensor):
        # x : shape ( batch, k, hidden_size)
        # annots : shape ( batch, seq_len, hidden_size)

        # -----
        # FILL THIS IN
        # Pay attention to which inputs you use for the queries, keys, and values.
        ↪ values.
        # -----
        # q =
        # k =
        # v =
```

```

q = self.Q(x)
k = self.K(annots)
v = self.V(annots)
return scaled_dot_attention(q, k, v)

```

1.2 Step 2: Causal Scaled Dot Product Attention

Using your `scaled_dot_attention` method, implement the `forward` method for the `CausalAttention` module.

```

[9]: class CausalAttention(Attention):
    def __init__(self, hidden_size: int):
        super().__init__(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        # x : shape ( batch, seq_len, hidden_size)

        # -----
        # FILL THIS IN
        # This should be very similar to your other attention implementation,
        # with the exception that you only have one set of outputs and you
        # must construct a causal attention mask.
        # -----
        # q =
        # k =
        # v =
        # mask =
        seq_len = x.size(1)
        q = self.Q(x)
        k = self.K(x)
        v = self.V(x)
        mask = torch.tril(torch.ones((seq_len, seq_len), device=x.device)).
        ↪ bool()
        return scaled_dot_attention(q, k, v, mask)

```

Simple test cases

Passing this DOES NOT mean your implementation is correct. But, if this fails, then you definitely did something wrong

```

[10]: cros_attn = Attention(32)
      caus_attn = CausalAttention(32)
      caus_attn.Q = cros_attn.Q
      caus_attn.K = cros_attn.K
      caus_attn.V = cros_attn.V

```

```

inp = torch.randn(4, 8, 32).to(device)
inp2 = torch.randn(4, 16, 32).to(device)

print("Checking consistent shape")
assert cros_attn(inp, inp2)[0].shape == inp.shape
assert caus_attn(inp)[0].shape == inp.shape
assert cros_attn(inp, inp2)[1].shape == (4, 8, 16)
assert caus_attn(inp)[1].shape == (4, 8, 8)

print("Checking attention mask")
assert torch.all(caus_attn(inp)[0][:, -1] == cros_attn(inp, inp)[0][:, -1])

print("Passed")

```

```

Checking consistent shape
Checking attention mask
Passed

```

Read through the following code. Make sure you understand what is going on. This code sets up the encoder-decoder architecture. It requires no additional code.

```

[11]: class PositionalEncoding(nn.Module):
    pe: torch.Tensor

    def __init__(self, hidden_size: int, max_len: int = 1000):
        super().__init__()

        pos = torch.arange(max_len).float().unsqueeze(1)
        dim = torch.arange(hidden_size // 2).float().unsqueeze(0)

        div_term = torch.exp(-math.log(10000.0) * (2 * dim) / hidden_size)
        angle = pos * div_term

        pe = torch.zeros(max_len, hidden_size)
        pe[:, 0::2] = torch.sin(angle)
        pe[:, 1::2] = torch.cos(angle)

        self.register_buffer("pe", pe)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size, max_len=max_len)

    def forward(self, idx: int):
        return self.pe[idx]

```

```

[12]: class MLP(nn.Module):

    def __init__(self, hidden_size: int):

```

```

        super().__init__()
        self.layer1 = nn.Linear(hidden_size, hidden_size, bias=False)
        self.layer2 = nn.Linear(hidden_size, hidden_size, bias=False)
        self.relu = nn.ReLU()
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        x = self.layer1(x)
        x = self.relu(x)
        x = self.layer2(x)
        return x

class EncoderBlock(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.norm1 = LayerNorm(hidden_size)
        self.norm2 = LayerNorm(hidden_size)
        self.attn = Attention(hidden_size)
        self.mlp = MLP(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        context, attention = self.attn(self.norm1(x), self.norm1(x))
        x = x + context
        x = x + self.mlp(self.norm2(x))
        return x, attention

class DecoderBlock(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.norm1 = LayerNorm(hidden_size)
        self.norm2 = LayerNorm(hidden_size)
        self.cros_attn = Attention(hidden_size)
        self.self_attn = CausalAttention(hidden_size)
        self.mlp = MLP(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor, annotation: torch.Tensor):
        context, self_attn = self.self_attn(self.norm1(x))
        x = x + context
        context, cros_attn = self.cros_attn(self.norm2(x), annotation)
        x = x + context

```

```

x = x + self.mlp(self.norm2(x))
return x, self_attn, cross_attn

```

```

[13]: class TransformerEncoderDecoder(nn.Module):
    def __init__(
        self, vocab_size: int, hidden_size: int, num_layers: int, dataset_name:
        ↪str
    ):
        super().__init__()

        self.tok_emb = nn.Embedding(vocab_size, hidden_size)
        self.pos_emb = PositionalEncoding(hidden_size)

        self.encoder_blocks = nn.ModuleList(
            [EncoderBlock(hidden_size) for i in range(num_layers)]
        )
        self.decoder_blocks = nn.ModuleList(
            [DecoderBlock(hidden_size) for i in range(num_layers)]
        )

        self.head = nn.Linear(hidden_size, vocab_size, bias=False)
        self.head.weight = self.tok_emb.weight
        # weight tie, the final layer reuses the weights of the embedding layers
        # this reduces the total number of parameters

        self.apply(self._init_weights) # initialize the weights to a small
        ↪number

        self.to(device)
        self.filename = model_filename(
            hidden_size=hidden_size,
            num_layers=num_layers,
            vocab_size=vocab_size,
            dataset_name=dataset_name,
        )

    def forward(self, inputs: torch.Tensor, annots: torch.Tensor):

        pos_x = torch.arange(
            0, inputs.shape[-1], device=inputs.device
        ).long() # creates tensor([0, 1, 2, 3, ...])
        pos_y = torch.arange(
            0, annots.shape[-1], device=inputs.device
        ).long() # creates tensor([0, 1, 2, 3, ...])

        x = self.tok_emb(inputs) + self.pos_emb(pos_x)
        y = self.tok_emb(annots) + self.pos_emb(pos_y)

```

```

self_attn = []
cros_attn = []

for encode in self.encoder_blocks:
    y, _ = encode(y)

for decode in self.decoder_blocks:
    x, s, c = decode(x, y)
    cros_attn.append(c)
    self_attn.append(s)

return self.head(x), {"self_attn": self_attn, "cros_attn": cros_attn}

def _init_weights(self, module: torch.nn.Module):
    if isinstance(module, nn.Linear):
        torch.nn.init.normal_(module.weight, mean=0.0, std=0.02)
        if module.bias is not None:
            torch.nn.init.zeros_(module.bias)
    elif isinstance(module, nn.Embedding):
        torch.nn.init.normal_(module.weight, mean=0.0, std=0.02)

```

1.3 Step 3: Training the model

You'll now train a set of example models. This section requires no additional code from you, with the exception of setting the datapath. However, **you are required to analyze and include some of its output.**

```

[14]: from dataset import EncoderDecoderDataset, DecoderDataset
      from utils import save_loss_comparison_by_dataset, □
      ↪ save_loss_comparison_by_hidden
      from torch.utils.data import DataLoader
      from trainer import train
      import torch.optim as optim

data_path = str("./") # Change data_path to where the data is stored

encdec_ds_small = EncoderDecoderDataset(data_path, "pig_latin_small")
encdec_ds_large = EncoderDecoderDataset(data_path, "pig_latin_large")

[15]: print("Dataset sample:")
      print("Input:", encdec_ds_small.to_token(encdec_ds_small[0][0].tolist()))
      print("Annotation:", encdec_ds_small.to_token(encdec_ds_small[0][1].tolist()))
      print("Target:", encdec_ds_small.to_token(encdec_ds_small[0][2].tolist()))

```

```

Dataset sample:
Input: <SOS>econcileray
Annotation: reconcile

```

Target: econcileray<EOS>

```
[16]: model = load_model_auto(
    TransformerEncoderDecoder,
    vocab_size=len(encdec_ds_small.idx_to_token),
    hidden_size=16,
    num_layers=4,
    dataset_name="small",
)
```

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid16_layers4.pt onto mps device.

```
[17]: optimizer = optim.AdamW(model.parameters(), lr=1e-2)
scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

results = train(
    model=model,
    dataset=encdec_ds_small,
    num_epochs=NUM_EPOCHS,
    optimizer=optimizer,
    scheduler=scheduler,
    device=device,
    batch_size=32,
)
save_model_auto(model)
```

```
Epoch 1 | Train loss: 1.551056 | Val loss 1.326135
Epoch 2 | Train loss: 1.063675 | Val loss 0.810254
Epoch 3 | Train loss: 0.601667 | Val loss 0.450644
Epoch 4 | Train loss: 0.395058 | Val loss 0.329814
Epoch 5 | Train loss: 0.339731 | Val loss 0.281942
Epoch 6 | Train loss: 0.242083 | Val loss 0.251930
Epoch 7 | Train loss: 0.201196 | Val loss 0.201325
Epoch 8 | Train loss: 0.165927 | Val loss 0.269903
Epoch 9 | Train loss: 0.156907 | Val loss 0.116667
Epoch 10 | Train loss: 0.119682 | Val loss 0.132721
Epoch 11 | Train loss: 0.111152 | Val loss 0.135230
Epoch 12 | Train loss: 0.096481 | Val loss 0.094605
Epoch 13 | Train loss: 0.089595 | Val loss 0.114751
Epoch 14 | Train loss: 0.087959 | Val loss 0.094765
Epoch 15 | Train loss: 0.077084 | Val loss 0.104997
Epoch 16 | Train loss: 0.061116 | Val loss 0.074258
Epoch 17 | Train loss: 0.064619 | Val loss 0.081565
Epoch 18 | Train loss: 0.063316 | Val loss 0.081086
Epoch 19 | Train loss: 0.045618 | Val loss 0.072637
Epoch 20 | Train loss: 0.051255 | Val loss 0.063279
Epoch 21 | Train loss: 0.041554 | Val loss 0.064884
```

```
Epoch 22 | Train loss: 0.035335 | Val loss 0.065694
Epoch 23 | Train loss: 0.035156 | Val loss 0.059162
Epoch 24 | Train loss: 0.036541 | Val loss 0.058218
Epoch 25 | Train loss: 0.034322 | Val loss 0.066668
```

```
[17]: PosixPath('/Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid1
6_layers4.pt')
```

```
[18]: def encdec_generate(
    input: str,
    model: TransformerEncoderDecoder,
    dataset: EncoderDecoderDataset,
    max_len: int = 20,
):
    gen_string = ""
    idx = torch.tensor([[dataset.start_idx]]).to(device)
    annots = torch.tensor(
        [dataset.to_idx(input) + [dataset.end_idx] * (max(0, 12 - len(input)))]
    ).to(device)
    # pad the input string with <EOS> since the collate_fn had to pad the
    ↪ stirngs

    for i in range(max_len):
        logits = model(idx, annots)[0]

        token_idx: torch.Tensor = logits[0, -1:].argmax(dim=-1)
        idx = torch.cat([idx, token_idx.unsqueeze(0)], dim=1)

        if token_idx.item() == dataset.end_idx:
            break
        else:
            gen_string = gen_string + dataset.idx_to_token[int(token_idx.
            ↪ item())]
    return gen_string

def encdec_translate(
    sentence: str, model: TransformerEncoderDecoder, dataset:
    ↪ EncoderDecoderDataset
):
    words = sentence.split()
    return " ".join([encdec_generate(w, model, dataset) for w in words])
```

```
[19]: encdec_translate("the", model, encdec_ds_small)
```

```
[19]: 'ethay'
```

```
[20]: encdec_translate("the air conditioning is working", model, encdec_ds_small)
```

```
[20]: 'ethay airway onditioningcay isway orkingway'
```

```
[21]: encdec_translate("testing some random words", model, encdec_ds_small)
```

```
[21]: 'estingtay omesay andomray ordsway'
```

```
[22]: configs = [  
    (16, "small", encdec_ds_small),  
    (16, "large", encdec_ds_large),  
    (32, "small", encdec_ds_small),  
    (32, "large", encdec_ds_large),  
]  
  
all_results = {}  
  
for hidden_size, dataset_name, dataset in configs:  
    print(f"\nTraining model: hidden_size={hidden_size},  
↪dataset={dataset_name}")  
  
    model = load_model_auto(  
        TransformerEncoderDecoder,  
        vocab_size=len(dataset.idx_to_token),  
        hidden_size=hidden_size,  
        num_layers=4,  
        dataset_name=dataset_name,  
    )  
  
    optimizer = optim.AdamW(model.parameters(), lr=1e-2)  
    scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)  
  
    results = train(  
        model=model,  
        dataset=dataset,  
        num_epochs=NUM_EPOCHS,  
        optimizer=optimizer,  
        scheduler=scheduler,  
        device=device,  
        batch_size=32 if dataset_name == "small" else 256,  
    )  
    all_results[(hidden_size, dataset_name)] = results  
    save_model_auto(model)  
  
save_loss_comparison_by_dataset(all_results)  
save_loss_comparison_by_hidden(all_results)
```

Training model: hidden_size=16, dataset=small

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid1_6_layers4.pt onto mps device.

Epoch 1		Train loss: 0.282690		Val loss 0.119487
Epoch 2		Train loss: 0.100203		Val loss 0.125501
Epoch 3		Train loss: 0.096975		Val loss 0.092090
Epoch 4		Train loss: 0.078260		Val loss 0.084708
Epoch 5		Train loss: 0.072440		Val loss 0.088962
Epoch 6		Train loss: 0.066571		Val loss 0.072962
Epoch 7		Train loss: 0.065810		Val loss 0.065583
Epoch 8		Train loss: 0.054039		Val loss 0.067781
Epoch 9		Train loss: 0.052774		Val loss 0.079471
Epoch 10		Train loss: 0.061673		Val loss 0.069800
Epoch 11		Train loss: 0.046289		Val loss 0.075188
Epoch 12		Train loss: 0.043467		Val loss 0.048739
Epoch 13		Train loss: 0.036280		Val loss 0.044235
Epoch 14		Train loss: 0.030028		Val loss 0.040429
Epoch 15		Train loss: 0.024368		Val loss 0.045205
Epoch 16		Train loss: 0.022556		Val loss 0.037809
Epoch 17		Train loss: 0.023777		Val loss 0.061744
Epoch 18		Train loss: 0.025582		Val loss 0.042941
Epoch 19		Train loss: 0.023655		Val loss 0.048473
Epoch 20		Train loss: 0.024986		Val loss 0.055363
Epoch 21		Train loss: 0.022103		Val loss 0.048486
Epoch 22		Train loss: 0.026233		Val loss 0.045333
Epoch 23		Train loss: 0.020105		Val loss 0.044494
Epoch 24		Train loss: 0.017601		Val loss 0.048049
Epoch 25		Train loss: 0.015755		Val loss 0.035441

Training model: hidden_size=16, dataset=large

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namelarge_vocab30_hid1_6_layers4.pt onto mps device.

Epoch 1		Train loss: 1.752035		Val loss 1.451140
Epoch 2		Train loss: 1.180100		Val loss 0.924659
Epoch 3		Train loss: 0.687570		Val loss 0.501195
Epoch 4		Train loss: 0.324483		Val loss 0.233330
Epoch 5		Train loss: 0.159920		Val loss 0.129188
Epoch 6		Train loss: 0.105429		Val loss 0.109710
Epoch 7		Train loss: 0.079490		Val loss 0.076954
Epoch 8		Train loss: 0.059346		Val loss 0.057958
Epoch 9		Train loss: 0.051220		Val loss 0.048695
Epoch 10		Train loss: 0.046706		Val loss 0.051182
Epoch 11		Train loss: 0.040899		Val loss 0.049593
Epoch 12		Train loss: 0.045514		Val loss 0.048623
Epoch 13		Train loss: 0.033743		Val loss 0.046426
Epoch 14		Train loss: 0.031032		Val loss 0.038175

Epoch 15		Train loss: 0.025308		Val loss 0.037642
Epoch 16		Train loss: 0.024815		Val loss 0.055294
Epoch 17		Train loss: 0.033792		Val loss 0.036898
Epoch 18		Train loss: 0.022894		Val loss 0.029995
Epoch 19		Train loss: 0.020354		Val loss 0.029650
Epoch 20		Train loss: 0.024549		Val loss 0.037606
Epoch 21		Train loss: 0.018849		Val loss 0.029841
Epoch 22		Train loss: 0.022382		Val loss 0.054181
Epoch 23		Train loss: 0.021724		Val loss 0.028102
Epoch 24		Train loss: 0.018688		Val loss 0.031005
Epoch 25		Train loss: 0.016014		Val loss 0.029370

Training model: hidden_size=32, dataset=small

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid3_2_layers4.pt onto mps device.

Epoch 1		Train loss: 1.908707		Val loss 1.791376
Epoch 2		Train loss: 1.339528		Val loss 1.073616
Epoch 3		Train loss: 0.691916		Val loss 0.417469
Epoch 4		Train loss: 0.329485		Val loss 0.226425
Epoch 5		Train loss: 0.215454		Val loss 0.190468
Epoch 6		Train loss: 0.166157		Val loss 0.140377
Epoch 7		Train loss: 0.130177		Val loss 0.110813
Epoch 8		Train loss: 0.093311		Val loss 0.092226
Epoch 9		Train loss: 0.088159		Val loss 0.123600
Epoch 10		Train loss: 0.088821		Val loss 0.061677
Epoch 11		Train loss: 0.061109		Val loss 0.071283
Epoch 12		Train loss: 0.047845		Val loss 0.063630
Epoch 13		Train loss: 0.046148		Val loss 0.052610
Epoch 14		Train loss: 0.042500		Val loss 0.066143
Epoch 15		Train loss: 0.040155		Val loss 0.044555
Epoch 16		Train loss: 0.032502		Val loss 0.050437
Epoch 17		Train loss: 0.026377		Val loss 0.038128
Epoch 18		Train loss: 0.025233		Val loss 0.052488
Epoch 19		Train loss: 0.027906		Val loss 0.036548
Epoch 20		Train loss: 0.016713		Val loss 0.039828
Epoch 21		Train loss: 0.018074		Val loss 0.038401
Epoch 22		Train loss: 0.014226		Val loss 0.032558
Epoch 23		Train loss: 0.018394		Val loss 0.037174
Epoch 24		Train loss: 0.017545		Val loss 0.034607
Epoch 25		Train loss: 0.013835		Val loss 0.032371

Training model: hidden_size=32, dataset=large

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namelarge_vocab30_hid3_2_layers4.pt onto mps device.

Epoch 1		Train loss: 1.869922		Val loss 1.322680
Epoch 2		Train loss: 0.861228		Val loss 0.365645

```

Epoch 3 | Train loss: 0.294439 | Val loss 0.231007
Epoch 4 | Train loss: 0.147985 | Val loss 0.088157
Epoch 5 | Train loss: 0.111792 | Val loss 0.125666
Epoch 6 | Train loss: 0.053795 | Val loss 0.049737
Epoch 7 | Train loss: 0.040531 | Val loss 0.108465
Epoch 8 | Train loss: 0.039519 | Val loss 0.047647
Epoch 9 | Train loss: 0.030697 | Val loss 0.033957
Epoch 10 | Train loss: 0.021484 | Val loss 0.069264
Epoch 11 | Train loss: 0.025629 | Val loss 0.032973
Epoch 12 | Train loss: 0.017451 | Val loss 0.022390
Epoch 13 | Train loss: 0.014279 | Val loss 0.022728
Epoch 14 | Train loss: 0.011467 | Val loss 0.018021
Epoch 15 | Train loss: 0.012247 | Val loss 0.038864
Epoch 16 | Train loss: 0.011687 | Val loss 0.016182
Epoch 17 | Train loss: 0.008074 | Val loss 0.015719
Epoch 18 | Train loss: 0.007249 | Val loss 0.021653
Epoch 19 | Train loss: 0.007246 | Val loss 0.020273
Epoch 20 | Train loss: 0.007894 | Val loss 0.018705
Epoch 21 | Train loss: 0.006733 | Val loss 0.015223
Epoch 22 | Train loss: 0.009007 | Val loss 0.030844
Epoch 23 | Train loss: 0.008515 | Val loss 0.026953
Epoch 24 | Train loss: 0.011375 | Val loss 0.016680
Epoch 25 | Train loss: 0.006743 | Val loss 0.019985

```

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<Figure size 640x480 with 0 Axes>

2 Part 2: Decoder-only Transformers and Multi-Head Attention

In this section, you'll train transformers that only use a decoder instead of a dual encoder-decoder architecture. You'll also implement and experiment with multi-head attention. Finally, you'll compare increasing parameter count with multi-head attention vs. increasing the parameter count with a larger hidden dimensionality.

2.1 Step 1: Implement the Decoder-Only Forward Method

```

[23]: class DecoderOnlyBlock(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.norm1 = LayerNorm(hidden_size)
        self.norm2 = LayerNorm(hidden_size)
        self.attn = CausalAttention(hidden_size)
        self.mlp = MLP(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):

```

```

        context, attention = self.attn(self.norm1(x))
        x = x + context
        x = x + self.mlp(self.norm2(x))
        return x, attention

class DecoderTransformer(nn.Module):
    def __init__(self, vocab_size: int, hidden_size: int, num_layers: int):
        super().__init__()

        self.tok_emb = nn.Embedding(vocab_size, hidden_size)
        self.pos_emb = PositionalEncoding(hidden_size)

        self.blocks = nn.ModuleList(
            [DecoderOnlyBlock(hidden_size) for i in range(num_layers)]
        )
        self.head = nn.Linear(hidden_size, vocab_size, bias=False)
        self.head.weight = self.tok_emb.weight
        # weight tie, the final layer reuses the weights of the embedding layers
        # this reduces the total number of parameters
        self.to(device)
        self.filename = model_filename(
            hidden_size=hidden_size, num_layers=num_layers,
            vocab_size=vocab_size
        )

    def forward(self, inputs: torch.Tensor):
        """Forward pass of the Transformer decoder.

        Arguments:
            inputs: Input token indexes across a batch for all the time step.
            (batch_size x decoder_seq_len)
        Returns:
            output: Un-normalized scores for each token in the vocabulary,
            across a batch for all the decoding time steps. (batch_size x
            decoder_seq_len x vocab_size)
            attentions: The stacked attention weights applied to the encoder
            annotations (batch_size x encoder_seq_len x decoder_seq_len)
        """
        # -----
        # FILL THIS IN
        # -----
        # pos =
        # x =
        # self_attn = []

        # for block in self.blocks:

```

```

#     ...
#
# unnormalized_scores =
pos = torch.arange(0, inputs.shape[-1], device=inputs.device).long()
x = self.tok_emb(inputs) + self.pos_emb(pos)
self_attn = []
for block in self.blocks:
    x, s = block(x)
    self_attn.append(s)
unnormalized_scores = self.head(x)

return unnormalized_scores, {"self_attn": self_attn}

```

```

[24]: decoder_ds_small = DecoderDataset(data_path, "pig_latin_small")
      decoder_ds_large = DecoderDataset(data_path, "pig_latin_large")

```

```

[25]: print("Dataset sample:")
      print("Input:", decoder_ds_small.to_token(decoder_ds_small[0][0].tolist()))
      print("Target:", decoder_ds_small.to_token(decoder_ds_small[0][1].tolist()))

```

Dataset sample:

Input: <SOS>reconcile<EOP>econcileray

Target: reconcile<EOP>econcileray<EOS>

2.2 Step 2: Format for decoder-only generation.

Implement the missing line in `decoder_generate` to create an input string that looks like:

<SOS> input sequence <EOP>

You should use the `dataset.start_idx`, `dataset.eop_idx` attributes and the `dataset.to_idx()` method.

```

[26]: def decoder_generate(
      input: str,
      model: DecoderTransformer,
      dataset: DecoderDataset,
      max_len: int = 20,
  ):
      gen_string = ""
      # idx = TODO
      idx = dataset.to_idx(input)
      idx = torch.tensor([idx]).to(device)

      for i in range(max_len):
          logits = model(idx)[0]

          token_idx = logits[0, -1:].argmax(dim=-1)
          idx = torch.cat([idx, token_idx.unsqueeze(0)], dim=1)

```

```

        if token_idx.item() == dataset.end_idx:
            break
        else:
            gen_string = gen_string + dataset.idx_to_token[token_idx.item()]
    return gen_string

def decoder_translate(
    sentence: str, model: DecoderTransformer, dataset: DecoderDataset
):
    words = sentence.split()
    return " ".join([decoder_generate(w, model, dataset) for w in words])

```

2.3 Step 3: Train the model

We'll now train a decoder only model and test it out on an example. Use the output of this to guide your answer for question 3.

```

[27]: model = load_model_auto(
        DecoderTransformer,
        vocab_size=len(decoder_ds_small.idx_to_token),
        hidden_size=16,
        num_layers=4,
    )

```

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/DecoderTransformer/Model_vocab30_hid16_layers4.pt onto mps device.

```

[28]: optimizer = optim.AdamW(model.parameters(), lr=1e-2)
        scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

        results = train(
            model=model,
            dataset=decoder_ds_small,
            num_epochs=NUM_EPOCHS,
            optimizer=optimizer,
            scheduler=scheduler,
            device=device,
            batch_size=32,
        )
        save_model_auto(model)

```

```

Epoch 1 | Train loss: 1.750797 | Val loss 1.543421
Epoch 2 | Train loss: 1.380129 | Val loss 1.224484
Epoch 3 | Train loss: 1.060866 | Val loss 0.986937
Epoch 4 | Train loss: 0.831978 | Val loss 0.764885
Epoch 5 | Train loss: 0.670749 | Val loss 0.790287

```

```
Epoch 6 | Train loss: 0.563655 | Val loss 0.504474
Epoch 7 | Train loss: 0.469036 | Val loss 0.553065
Epoch 8 | Train loss: 0.384653 | Val loss 0.436691
Epoch 9 | Train loss: 0.340577 | Val loss 0.286533
Epoch 10 | Train loss: 0.291354 | Val loss 0.356906
Epoch 11 | Train loss: 0.275355 | Val loss 0.218699
Epoch 12 | Train loss: 0.235852 | Val loss 0.234852
Epoch 13 | Train loss: 0.207349 | Val loss 0.194345
Epoch 14 | Train loss: 0.202178 | Val loss 0.226432
Epoch 15 | Train loss: 0.181114 | Val loss 0.174726
Epoch 16 | Train loss: 0.160138 | Val loss 0.167639
Epoch 17 | Train loss: 0.151384 | Val loss 0.125920
Epoch 18 | Train loss: 0.141745 | Val loss 0.153938
Epoch 19 | Train loss: 0.149035 | Val loss 0.134451
Epoch 20 | Train loss: 0.124795 | Val loss 0.127090
Epoch 21 | Train loss: 0.104896 | Val loss 0.123745
Epoch 22 | Train loss: 0.104194 | Val loss 0.109657
Epoch 23 | Train loss: 0.100252 | Val loss 0.142833
Epoch 24 | Train loss: 0.098598 | Val loss 0.119624
Epoch 25 | Train loss: 0.091952 | Val loss 0.105700
```

```
[28]: PosixPath('/Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/DecoderTransformer/Model_vocab30_hid16_layers4.pt')
```

```
[29]: decoder_translate("the air conditioning is working", model, decoder_ds_small)
```

```
[29]: 'fohefohgggggggggggggg gggrgrgggggggggggggg bondccioniningondunr
ssssssssssssssssssss rlh'
```

2.4 Step 4: Implement Multi-Head Attention

In this next step, you'll implement multi-head attention. In this setting, we'll have the dimensionality for each of the heads be $(\text{total hidden dimensionality})/(\text{number of heads})$. We'll use a projection head to combine the outputs of these different attention heads after concatenating their outputs together (which can also be achieved using tensor reshaping).

```
[30]: # TO DO:

def multihead_scaled_dot_attention(
    q: torch.Tensor, k: torch.Tensor, v: torch.Tensor, mask: torch.Tensor | None = None
):
    """Compute multi-head dot product attention.
    q : shape (batch, num_heads, k, hidden_size)
    k : shape (batch, num_heads, seq_len, hidden_size)
    v : shape (batch, num_heads, seq_len, hidden_size)
```

```

    returns:
        context : shape (batch, num_heads, k, hidden_size)
        attention : shape (batch, num_heads, k, seq_len)
    """
    # Hint: if you implemented scaled_dot_attention well, the code for this
    ↪ should be the exact same.

    # -----
    # FILL THIS IN
    # -----
    # unnorm_attn =
    # masked_unnorm_attn =
    # attention =
    # context =
    unnorm_attn = torch.matmul(q, k.transpose(-2, -1)) / math.sqrt(q.size(-1))
    if mask is not None:
        unnorm_attn = unnorm_attn.masked_fill(mask == 0, float("-inf"))
    attention = F.softmax(unnorm_attn, dim=-1)
    context = torch.matmul(attention, v)
    return context, attention

```

```

[31]: class MultiheadCausalAttention(Attention):
    proj: nn.Linear

    def __init__(self, hidden_size: int, num_heads: int):
        super().__init__(hidden_size)
        # self.Q, self.K, self.V already defined as nn.Linear(hidden_size,
    ↪ hidden_size)
        self.num_heads = num_heads
        # -----
        # FILL THIS IN
        # -----
        # if self.num_heads > 1:
        #     self.proj = nn.Linear( , , bias=False)
        if self.num_heads > 1:
            self.proj = nn.Linear(hidden_size, hidden_size, bias=False)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size,
    ↪ num_heads=num_heads)

    def forward(self, x: torch.Tensor):
        """Compute causal multi-head dot product attention from an input
    ↪ sequence.
        x : shape ( batch, seq_len, hidden_size)
        k : shape (batch, num_heads, seq_len, hidden_size)
        v : shape (batch, num_heads, seq_len, hidden_size)

```

```

    returns:
        context : shape ( batch, seq_len, hidden_size)
        attention : shape (batch, num_heads, k, seq_len)
    """
    # -----
    # FILL THIS IN
    # -----
    # unnorm_attn =
    # masked_unnorm_attn =
    # attention =
    # context =

    # B, T, C = x.size()
    # q = self.Q(x).view( , , , ).transpose(1, 2)
    # k = self.K(x).view( , , , ).transpose(1, 2)
    # v = self.V(x).view( , , , ).transpose(1, 2)

    # mask =

    # context, attention = multihead_scaled_dot_attention(q, k, v, mask)

    # Hint: you may find the .transpose(), .contiguous(), and .view()
    # methods helpful for creating the reshaped context.

    # reshaped_context =

    # Only perform a projection if the number of heads is more than 1.
    B, T, C = x.size()
    q = self.Q(x).view(B, T, self.num_heads, C // self.num_heads).
    ↪transpose(1, 2)
    k = self.K(x).view(B, T, self.num_heads, C // self.num_heads).
    ↪transpose(1, 2)
    v = self.V(x).view(B, T, self.num_heads, C // self.num_heads).
    ↪transpose(1, 2)
    mask = torch.tril(torch.ones((T, T), device=x.device)).bool()
    context, attention = multihead_scaled_dot_attention(q, k, v, mask)
    reshaped_context = context.transpose(1, 2).contiguous().view(B, T, C)
    if self.num_heads > 1:

        return self.proj(reshaped_context), attention

    else:

        return reshaped_context, attention

```

```

[32]: class MultiheadDecoderBlock(DecoderOnlyBlock):
        def __init__(self, hidden_size: int, num_heads: int):

```

```

        super().__init__(hidden_size)
        self.attn = MultiheadCausalAttention(hidden_size, num_heads)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size,
↪num_heads=num_heads)

class MultiheadDecoderTransformer(DecoderTransformer):
    def __init__(
        self, vocab_size: int, hidden_size: int, num_layers: int, num_heads: int
    ):
        super().__init__(vocab_size, hidden_size, num_layers)
        self.num_heads = num_heads
        self.num_layers = num_layers
        self.vocab_size = vocab_size
        self.hidden_size = hidden_size

        self.blocks = nn.ModuleList(
            [MultiheadDecoderBlock(hidden_size, num_heads) for i in
↪range(num_layers)]
        )
        self.to(device)
        self.filename = model_filename(
            hidden_size=hidden_size,
            num_layers=num_layers,
            vocab_size=vocab_size,
            num_heads=num_heads,
        )

```

2.5 Step 5: Compare Parameter Usage Methods

Here, you'll train 5 models with varying numbers of attention heads and hidden dimensionalities. You'll compare their parameter efficiency. Start by implementing the `calc_parameters_decoder` method, and then run the code to train the models.

```

[33]: def calc_parameters_decoder(model: MultiheadDecoderTransformer) -> int:
    # -----
    # FILL THIS IN
    # -----
    # params = TODO # embedding
    # params += TODO # QKV linear projections
    # if model.num_heads > 1:
    #     params += TODO # attention linear projection
    # params += TODO # MLP linear
    # params += TODO # layer norms Hint: LayerNorm(x) uses x parameters

    # Hint: Read the code carefully:

```

```

    # params += TODO # head
    params = model.vocab_size * model.hidden_size # embedding
    params += 3 * (model.hidden_size * model.hidden_size) # QKV linear
    ↪projections
    if model.num_heads > 1:
        params += model.hidden_size * model.hidden_size # attention linear
    ↪projection
    params += 2 * (
        4 * model.hidden_size * model.hidden_size
    ) # MLP linear (2 per block)
    params += 2 * 2 * model.hidden_size # layer norms (2 per block)

    return params

```

```

[34]: import matplotlib.pyplot as plt
import numpy as np

configs = [
    (16, 4, 4, 64), # (hidden_size, num_heads, num_layers, batch_size)
    (16, 2, 4, 64),
    (16, 1, 4, 64),
    (24, 1, 4, 64),
    (32, 1, 4, 64),
]

all_results = {}

for hidden_size, num_heads, num_layers, batch_size in configs:
    print(
        f"\nTraining model: hidden_size={hidden_size}, num_heads={num_heads},
    ↪num_layers={num_layers}"
    )

    model = load_model_auto(
        MultiheadDecoderTransformer,
        vocab_size=len(decoder_ds_large.idx_to_token),
        hidden_size=hidden_size,
        num_layers=num_layers,
        num_heads=num_heads,
    )

    optimizer = optim.AdamW(model.parameters(), lr=5e-3)
    scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

    results = train(
        model=model,
        dataset=decoder_ds_large,

```

```

        num_epochs=NUM_EPOCHS,
        optimizer=optimizer,
        scheduler=scheduler,
        device=device,
        batch_size=batch_size,
    )
    save_model_auto(model)

    parameters = calc_parameters_decoder(model)
    x_points = np.linspace(0, len(results["val_loss"]),
    ↪len(results["val_loss"]))
    plt.plot(
        x_points,
        results["val_loss"],
        label="heads = {}, dim={}, params={}".format(
            num_heads, hidden_size, parameters
        ),
    )

plt.xlabel("Epochs")
plt.ylabel("Validation Loss")
plt.title("Compute Optimal Models")
plt.grid()
plt.legend()
plt.show()

```

Training model: hidden_size=16, num_heads=4, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads4.pt onto mps device.

Epoch 1	Train loss: 0.813669	Val loss 0.456453
Epoch 2	Train loss: 0.318358	Val loss 0.188327
Epoch 3	Train loss: 0.181642	Val loss 0.152886
Epoch 4	Train loss: 0.124241	Val loss 0.086995
Epoch 5	Train loss: 0.096061	Val loss 0.076589
Epoch 6	Train loss: 0.085140	Val loss 0.106006
Epoch 7	Train loss: 0.074020	Val loss 0.058294
Epoch 8	Train loss: 0.060551	Val loss 0.082235
Epoch 9	Train loss: 0.053440	Val loss 0.055254
Epoch 10	Train loss: 0.046854	Val loss 0.052933
Epoch 11	Train loss: 0.039619	Val loss 0.037227
Epoch 12	Train loss: 0.036588	Val loss 0.040144
Epoch 13	Train loss: 0.031300	Val loss 0.030104
Epoch 14	Train loss: 0.029731	Val loss 0.032388
Epoch 15	Train loss: 0.025695	Val loss 0.031068
Epoch 16	Train loss: 0.025639	Val loss 0.030745
Epoch 17	Train loss: 0.023204	Val loss 0.023949

Epoch 18		Train loss: 0.018776		Val loss 0.020600
Epoch 19		Train loss: 0.020193		Val loss 0.031106
Epoch 20		Train loss: 0.018875		Val loss 0.025123
Epoch 21		Train loss: 0.016650		Val loss 0.019878
Epoch 22		Train loss: 0.015947		Val loss 0.019585
Epoch 23		Train loss: 0.016648		Val loss 0.025075
Epoch 24		Train loss: 0.013225		Val loss 0.025776
Epoch 25		Train loss: 0.013089		Val loss 0.018158

Training model: hidden_size=16, num_heads=2, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads2.pt onto mps device.

Epoch 1		Train loss: 1.018973		Val loss 0.788565
Epoch 2		Train loss: 0.500485		Val loss 0.370172
Epoch 3		Train loss: 0.265157		Val loss 0.281128
Epoch 4		Train loss: 0.187128		Val loss 0.176626
Epoch 5		Train loss: 0.143767		Val loss 0.128517
Epoch 6		Train loss: 0.114133		Val loss 0.076675
Epoch 7		Train loss: 0.095616		Val loss 0.068469
Epoch 8		Train loss: 0.082454		Val loss 0.065698
Epoch 9		Train loss: 0.079078		Val loss 0.058812
Epoch 10		Train loss: 0.061667		Val loss 0.080626
Epoch 11		Train loss: 0.059266		Val loss 0.142069
Epoch 12		Train loss: 0.050418		Val loss 0.045068
Epoch 13		Train loss: 0.048671		Val loss 0.088596
Epoch 14		Train loss: 0.037041		Val loss 0.038197
Epoch 15		Train loss: 0.040018		Val loss 0.058965
Epoch 16		Train loss: 0.031405		Val loss 0.028674
Epoch 17		Train loss: 0.028455		Val loss 0.033782
Epoch 18		Train loss: 0.029210		Val loss 0.027892
Epoch 19		Train loss: 0.024781		Val loss 0.037200
Epoch 20		Train loss: 0.025257		Val loss 0.027041
Epoch 21		Train loss: 0.023607		Val loss 0.029257
Epoch 22		Train loss: 0.020624		Val loss 0.028002
Epoch 23		Train loss: 0.020528		Val loss 0.025124
Epoch 24		Train loss: 0.018870		Val loss 0.023551
Epoch 25		Train loss: 0.018044		Val loss 0.030579

Training model: hidden_size=16, num_heads=1, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads1.pt onto mps device.

Epoch 1		Train loss: 0.745471		Val loss 0.481261
Epoch 2		Train loss: 0.438703		Val loss 0.362150
Epoch 3		Train loss: 0.301155		Val loss 0.225376
Epoch 4		Train loss: 0.224585		Val loss 0.164387
Epoch 5		Train loss: 0.193241		Val loss 0.159801

Epoch 6	Train loss: 0.156916	Val loss 0.126491
Epoch 7	Train loss: 0.138336	Val loss 0.128567
Epoch 8	Train loss: 0.118482	Val loss 0.095112
Epoch 9	Train loss: 0.115696	Val loss 0.116222
Epoch 10	Train loss: 0.100234	Val loss 0.137132
Epoch 11	Train loss: 0.091130	Val loss 0.072352
Epoch 12	Train loss: 0.078085	Val loss 0.077328
Epoch 13	Train loss: 0.081894	Val loss 0.066050
Epoch 14	Train loss: 0.063385	Val loss 0.064379
Epoch 15	Train loss: 0.066233	Val loss 0.054153
Epoch 16	Train loss: 0.064165	Val loss 0.074182
Epoch 17	Train loss: 0.054361	Val loss 0.058246
Epoch 18	Train loss: 0.051486	Val loss 0.053706
Epoch 19	Train loss: 0.047319	Val loss 0.045460
Epoch 20	Train loss: 0.047748	Val loss 0.050424
Epoch 21	Train loss: 0.043327	Val loss 0.049133
Epoch 22	Train loss: 0.038748	Val loss 0.099252
Epoch 23	Train loss: 0.037680	Val loss 0.036589
Epoch 24	Train loss: 0.036105	Val loss 0.039024
Epoch 25	Train loss: 0.035018	Val loss 0.045957

Training model: hidden_size=24, num_heads=1, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid24_layers4_heads1.pt onto mps device.

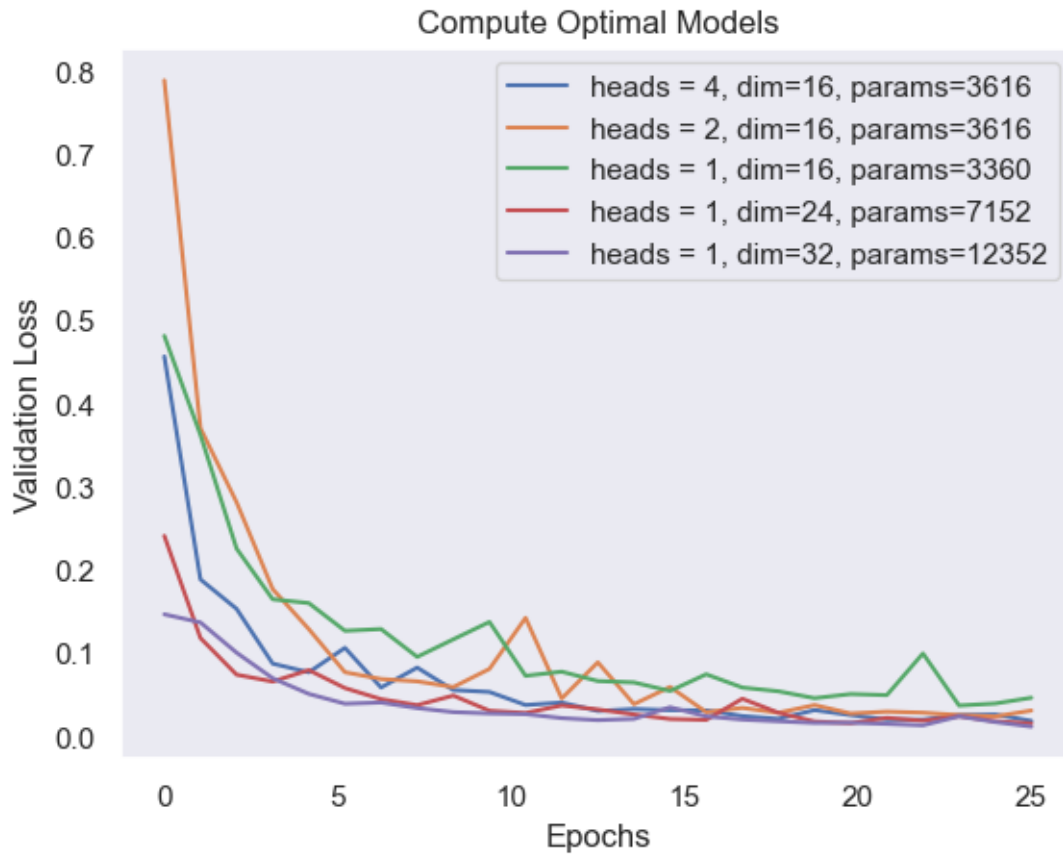
Epoch 1	Train loss: 0.382138	Val loss 0.240671
Epoch 2	Train loss: 0.148967	Val loss 0.117768
Epoch 3	Train loss: 0.095751	Val loss 0.073707
Epoch 4	Train loss: 0.076854	Val loss 0.065455
Epoch 5	Train loss: 0.065960	Val loss 0.079231
Epoch 6	Train loss: 0.050421	Val loss 0.057626
Epoch 7	Train loss: 0.044531	Val loss 0.044251
Epoch 8	Train loss: 0.036336	Val loss 0.037041
Epoch 9	Train loss: 0.033996	Val loss 0.048603
Epoch 10	Train loss: 0.029491	Val loss 0.030315
Epoch 11	Train loss: 0.027907	Val loss 0.027635
Epoch 12	Train loss: 0.023680	Val loss 0.036719
Epoch 13	Train loss: 0.024870	Val loss 0.031910
Epoch 14	Train loss: 0.018978	Val loss 0.025906
Epoch 15	Train loss: 0.020195	Val loss 0.020388
Epoch 16	Train loss: 0.018747	Val loss 0.019331
Epoch 17	Train loss: 0.015067	Val loss 0.044780
Epoch 18	Train loss: 0.015846	Val loss 0.027907
Epoch 19	Train loss: 0.013938	Val loss 0.017379
Epoch 20	Train loss: 0.013059	Val loss 0.015499
Epoch 21	Train loss: 0.011967	Val loss 0.021743
Epoch 22	Train loss: 0.011263	Val loss 0.018424
Epoch 23	Train loss: 0.010913	Val loss 0.023736

Epoch 24 | Train loss: 0.010483 | Val loss 0.017777
Epoch 25 | Train loss: 0.009405 | Val loss 0.014990

Training model: hidden_size=32, num_heads=1, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid32_layers4_heads1.pt onto mps device.

Epoch 1 | Train loss: 0.286629 | Val loss 0.146302
Epoch 2 | Train loss: 0.125485 | Val loss 0.136549
Epoch 3 | Train loss: 0.095803 | Val loss 0.100186
Epoch 4 | Train loss: 0.069927 | Val loss 0.069415
Epoch 5 | Train loss: 0.059556 | Val loss 0.050596
Epoch 6 | Train loss: 0.043010 | Val loss 0.038776
Epoch 7 | Train loss: 0.035310 | Val loss 0.040229
Epoch 8 | Train loss: 0.030662 | Val loss 0.033544
Epoch 9 | Train loss: 0.026778 | Val loss 0.028711
Epoch 10 | Train loss: 0.025704 | Val loss 0.027127
Epoch 11 | Train loss: 0.023963 | Val loss 0.026332
Epoch 12 | Train loss: 0.018128 | Val loss 0.021484
Epoch 13 | Train loss: 0.017573 | Val loss 0.019068
Epoch 14 | Train loss: 0.015893 | Val loss 0.020552
Epoch 15 | Train loss: 0.015352 | Val loss 0.034526
Epoch 16 | Train loss: 0.012132 | Val loss 0.023576
Epoch 17 | Train loss: 0.011141 | Val loss 0.019705
Epoch 18 | Train loss: 0.012571 | Val loss 0.017661
Epoch 19 | Train loss: 0.009691 | Val loss 0.015707
Epoch 20 | Train loss: 0.007921 | Val loss 0.015915
Epoch 21 | Train loss: 0.008482 | Val loss 0.014529
Epoch 22 | Train loss: 0.007229 | Val loss 0.012837
Epoch 23 | Train loss: 0.008125 | Val loss 0.023578
Epoch 24 | Train loss: 0.007693 | Val loss 0.016775
Epoch 25 | Train loss: 0.003847 | Val loss 0.011257



```
[35]: plt.clf()
```

<Figure size 640x480 with 0 Axes>

3 Part 3: IsoFLOP Profiling and Scaling Laws

In this final section, you'll compare different architectures based on their compute efficiency. You will also determine the optimal number of parameters and tokens for a fixed compute budget.

While this section does not require any additional code from you, **use the output of it to guide your written responses.**

```
[ ]: from utils import (
    plot_scaling_law,
    plot_scaling_law_poly,
    interpolate,
    plot_flops_tokens,
    plot_isoflop,
    plot_flops_params,
)
```

```
import numpy as np
```

3.1 Step 1: Train models, Compare FLOPS vs Validation Loss

We'll inspect the compute efficiency of different models by plotting their validation loss against the number of FLOPs used in training up to that point.

```
[37]: from types_hw2 import TrainerHistDict
```

```
[38]: configs = [
    (40, 2, 4, 256), # (hidden_size, num_heads, num_layers, batch_size)
    (32, 2, 4, 128),
    (24, 2, 4, 64),
    (16, 1, 4, 64),
    (16, 1, 3, 64),
    (12, 1, 3, 64),
    (12, 1, 2, 64),
]

all_results: dict[int, TrainerHistDict] = {}

for hidden_size, num_heads, num_layers, batch_size in configs:
    print(
        f"\nTraining model: hidden_size={hidden_size}, num_heads={num_heads},
        ↪num_layers={num_layers}"
    )

    model = load_model_auto(
        MultiheadDecoderTransformer,
        vocab_size=len(decoder_ds_large.idx_to_token),
        hidden_size=hidden_size,
        num_layers=num_layers,
        num_heads=num_heads,
    )

    optimizer = optim.AdamW(model.parameters(), lr=5e-3)
    scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

    results = train(
        model=model,
        dataset=decoder_ds_large,
        num_epochs=NUM_EPOCHS,
        optimizer=optimizer,
        scheduler=scheduler,
        device=device,
        batch_size=batch_size,
```

```
)
save_model_auto(model)

parameters = calc_parameters_decoder(model)
all_results[parameters] = results
```

Training model: hidden_size=40, num_heads=2, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid40_layers4_heads2.pt onto mps device.

Epoch 1		Train loss: 1.528885		Val loss 0.670217
Epoch 2		Train loss: 0.346934		Val loss 0.153018
Epoch 3		Train loss: 0.123261		Val loss 0.108073
Epoch 4		Train loss: 0.062645		Val loss 0.078309
Epoch 5		Train loss: 0.047663		Val loss 0.045472
Epoch 6		Train loss: 0.031922		Val loss 0.052582
Epoch 7		Train loss: 0.033140		Val loss 0.030728
Epoch 8		Train loss: 0.020986		Val loss 0.028729
Epoch 9		Train loss: 0.017214		Val loss 0.024835
Epoch 10		Train loss: 0.018561		Val loss 0.038733
Epoch 11		Train loss: 0.017493		Val loss 0.020770
Epoch 12		Train loss: 0.008237		Val loss 0.017349
Epoch 13		Train loss: 0.005149		Val loss 0.013267
Epoch 14		Train loss: 0.006774		Val loss 0.019120
Epoch 15		Train loss: 0.007481		Val loss 0.014373
Epoch 16		Train loss: 0.003869		Val loss 0.015374
Epoch 17		Train loss: 0.003134		Val loss 0.026414
Epoch 18		Train loss: 0.004332		Val loss 0.015742
Epoch 19		Train loss: 0.004408		Val loss 0.016294
Epoch 20		Train loss: 0.002298		Val loss 0.011664
Epoch 21		Train loss: 0.002010		Val loss 0.011788
Epoch 22		Train loss: 0.001391		Val loss 0.010979
Epoch 23		Train loss: 0.001266		Val loss 0.010547
Epoch 24		Train loss: 0.000736		Val loss 0.010512
Epoch 25		Train loss: 0.000582		Val loss 0.010569

Training model: hidden_size=32, num_heads=2, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid32_layers4_heads2.pt onto mps device.

Epoch 1		Train loss: 0.626203		Val loss 0.226660
Epoch 2		Train loss: 0.158771		Val loss 0.111884
Epoch 3		Train loss: 0.091994		Val loss 0.078803
Epoch 4		Train loss: 0.067058		Val loss 0.093248
Epoch 5		Train loss: 0.042171		Val loss 0.050964
Epoch 6		Train loss: 0.034123		Val loss 0.040262
Epoch 7		Train loss: 0.026373		Val loss 0.034361

Epoch 8	Train loss: 0.022235	Val loss 0.037488
Epoch 9	Train loss: 0.020471	Val loss 0.023280
Epoch 10	Train loss: 0.017970	Val loss 0.020324
Epoch 11	Train loss: 0.015300	Val loss 0.020693
Epoch 12	Train loss: 0.010333	Val loss 0.014092
Epoch 13	Train loss: 0.006809	Val loss 0.013733
Epoch 14	Train loss: 0.009267	Val loss 0.016764
Epoch 15	Train loss: 0.008585	Val loss 0.012869
Epoch 16	Train loss: 0.007558	Val loss 0.016300
Epoch 17	Train loss: 0.004313	Val loss 0.010345
Epoch 18	Train loss: 0.004013	Val loss 0.012181
Epoch 19	Train loss: 0.004019	Val loss 0.009493
Epoch 20	Train loss: 0.003804	Val loss 0.011114
Epoch 21	Train loss: 0.003044	Val loss 0.015951
Epoch 22	Train loss: 0.004143	Val loss 0.013995
Epoch 23	Train loss: 0.004601	Val loss 0.013356
Epoch 24	Train loss: 0.002478	Val loss 0.008538
Epoch 25	Train loss: 0.003044	Val loss 0.013154

Training model: hidden_size=24, num_heads=2, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid24_layers4_heads2.pt onto mps device.

Epoch 1	Train loss: 0.332824	Val loss 0.137187
Epoch 2	Train loss: 0.136318	Val loss 0.102007
Epoch 3	Train loss: 0.098672	Val loss 0.096196
Epoch 4	Train loss: 0.077745	Val loss 0.080298
Epoch 5	Train loss: 0.061763	Val loss 0.059715
Epoch 6	Train loss: 0.048399	Val loss 0.050352
Epoch 7	Train loss: 0.044509	Val loss 0.035286
Epoch 8	Train loss: 0.037407	Val loss 0.039287
Epoch 9	Train loss: 0.032497	Val loss 0.031630
Epoch 10	Train loss: 0.027827	Val loss 0.045814
Epoch 11	Train loss: 0.026918	Val loss 0.024948
Epoch 12	Train loss: 0.022797	Val loss 0.038503
Epoch 13	Train loss: 0.021761	Val loss 0.024778
Epoch 14	Train loss: 0.018936	Val loss 0.023858
Epoch 15	Train loss: 0.017815	Val loss 0.027666
Epoch 16	Train loss: 0.017429	Val loss 0.023673
Epoch 17	Train loss: 0.015927	Val loss 0.017590
Epoch 18	Train loss: 0.013313	Val loss 0.039375
Epoch 19	Train loss: 0.014372	Val loss 0.019983
Epoch 20	Train loss: 0.012826	Val loss 0.017868
Epoch 21	Train loss: 0.011387	Val loss 0.018590
Epoch 22	Train loss: 0.010276	Val loss 0.022637
Epoch 23	Train loss: 0.009457	Val loss 0.012275
Epoch 24	Train loss: 0.008262	Val loss 0.013928
Epoch 25	Train loss: 0.007551	Val loss 0.013872

Training model: hidden_size=16, num_heads=1, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads1.pt onto mps device.

Epoch 1		Train loss: 0.298407		Val loss 0.130236
Epoch 2		Train loss: 0.084526		Val loss 0.075383
Epoch 3		Train loss: 0.073439		Val loss 0.072686
Epoch 4		Train loss: 0.074884		Val loss 0.318166
Epoch 5		Train loss: 0.065450		Val loss 0.056413
Epoch 6		Train loss: 0.062733		Val loss 0.046585
Epoch 7		Train loss: 0.057245		Val loss 0.046918
Epoch 8		Train loss: 0.051880		Val loss 0.058239
Epoch 9		Train loss: 0.049230		Val loss 0.066891
Epoch 10		Train loss: 0.043765		Val loss 0.094727
Epoch 11		Train loss: 0.042195		Val loss 0.042310
Epoch 12		Train loss: 0.036860		Val loss 0.048052
Epoch 13		Train loss: 0.034815		Val loss 0.041075
Epoch 14		Train loss: 0.033059		Val loss 0.057601
Epoch 15		Train loss: 0.031777		Val loss 0.039322
Epoch 16		Train loss: 0.031798		Val loss 0.036972
Epoch 17		Train loss: 0.029104		Val loss 0.043511
Epoch 18		Train loss: 0.028445		Val loss 0.033316
Epoch 19		Train loss: 0.025779		Val loss 0.032366
Epoch 20		Train loss: 0.023755		Val loss 0.027126
Epoch 21		Train loss: 0.021677		Val loss 0.033200
Epoch 22		Train loss: 0.021191		Val loss 0.028775
Epoch 23		Train loss: 0.020917		Val loss 0.028452
Epoch 24		Train loss: 0.020547		Val loss 0.027248
Epoch 25		Train loss: 0.018630		Val loss 0.022767

Training model: hidden_size=16, num_heads=1, num_layers=3

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers3_heads1.pt onto mps device.

Epoch 1		Train loss: 1.068045		Val loss 0.781454
Epoch 2		Train loss: 0.550112		Val loss 0.421194
Epoch 3		Train loss: 0.374141		Val loss 0.313414
Epoch 4		Train loss: 0.287023		Val loss 0.305681
Epoch 5		Train loss: 0.236598		Val loss 0.272777
Epoch 6		Train loss: 0.198353		Val loss 0.172080
Epoch 7		Train loss: 0.172094		Val loss 0.170062
Epoch 8		Train loss: 0.156113		Val loss 0.154789
Epoch 9		Train loss: 0.137031		Val loss 0.122783
Epoch 10		Train loss: 0.124263		Val loss 0.128503
Epoch 11		Train loss: 0.110700		Val loss 0.115411
Epoch 12		Train loss: 0.102132		Val loss 0.108959
Epoch 13		Train loss: 0.091151		Val loss 0.089639

Epoch 14		Train loss: 0.086317		Val loss 0.080951
Epoch 15		Train loss: 0.080013		Val loss 0.110410
Epoch 16		Train loss: 0.076733		Val loss 0.076909
Epoch 17		Train loss: 0.070704		Val loss 0.071879
Epoch 18		Train loss: 0.063811		Val loss 0.075305
Epoch 19		Train loss: 0.061596		Val loss 0.070365
Epoch 20		Train loss: 0.058951		Val loss 0.096296
Epoch 21		Train loss: 0.054408		Val loss 0.061742
Epoch 22		Train loss: 0.050902		Val loss 0.059201
Epoch 23		Train loss: 0.048161		Val loss 0.099926
Epoch 24		Train loss: 0.046163		Val loss 0.058247
Epoch 25		Train loss: 0.045451		Val loss 0.064608

Training model: hidden_size=12, num_heads=1, num_layers=3

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid12_layers3_heads1.pt onto mps device.

Epoch 1		Train loss: 1.606250		Val loss 1.486954
Epoch 2		Train loss: 1.340124		Val loss 1.207574
Epoch 3		Train loss: 1.038600		Val loss 0.861892
Epoch 4		Train loss: 0.782728		Val loss 0.707117
Epoch 5		Train loss: 0.634717		Val loss 0.558059
Epoch 6		Train loss: 0.537351		Val loss 0.553718
Epoch 7		Train loss: 0.479854		Val loss 0.460927
Epoch 8		Train loss: 0.434182		Val loss 0.491324
Epoch 9		Train loss: 0.417915		Val loss 0.442564
Epoch 10		Train loss: 0.376191		Val loss 0.378826
Epoch 11		Train loss: 0.341910		Val loss 0.331617
Epoch 12		Train loss: 0.324967		Val loss 0.320721
Epoch 13		Train loss: 0.302648		Val loss 0.366901
Epoch 14		Train loss: 0.288819		Val loss 0.289654
Epoch 15		Train loss: 0.272418		Val loss 0.285745
Epoch 16		Train loss: 0.259177		Val loss 0.251023
Epoch 17		Train loss: 0.241572		Val loss 0.229630
Epoch 18		Train loss: 0.225692		Val loss 0.279837
Epoch 19		Train loss: 0.215638		Val loss 0.230677
Epoch 20		Train loss: 0.205868		Val loss 0.199519
Epoch 21		Train loss: 0.201496		Val loss 0.212249
Epoch 22		Train loss: 0.183242		Val loss 0.180133
Epoch 23		Train loss: 0.175283		Val loss 0.235717
Epoch 24		Train loss: 0.170182		Val loss 0.174715
Epoch 25		Train loss: 0.164392		Val loss 0.159902

Training model: hidden_size=12, num_heads=1, num_layers=2

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid12_layers2_heads1.pt onto mps device.

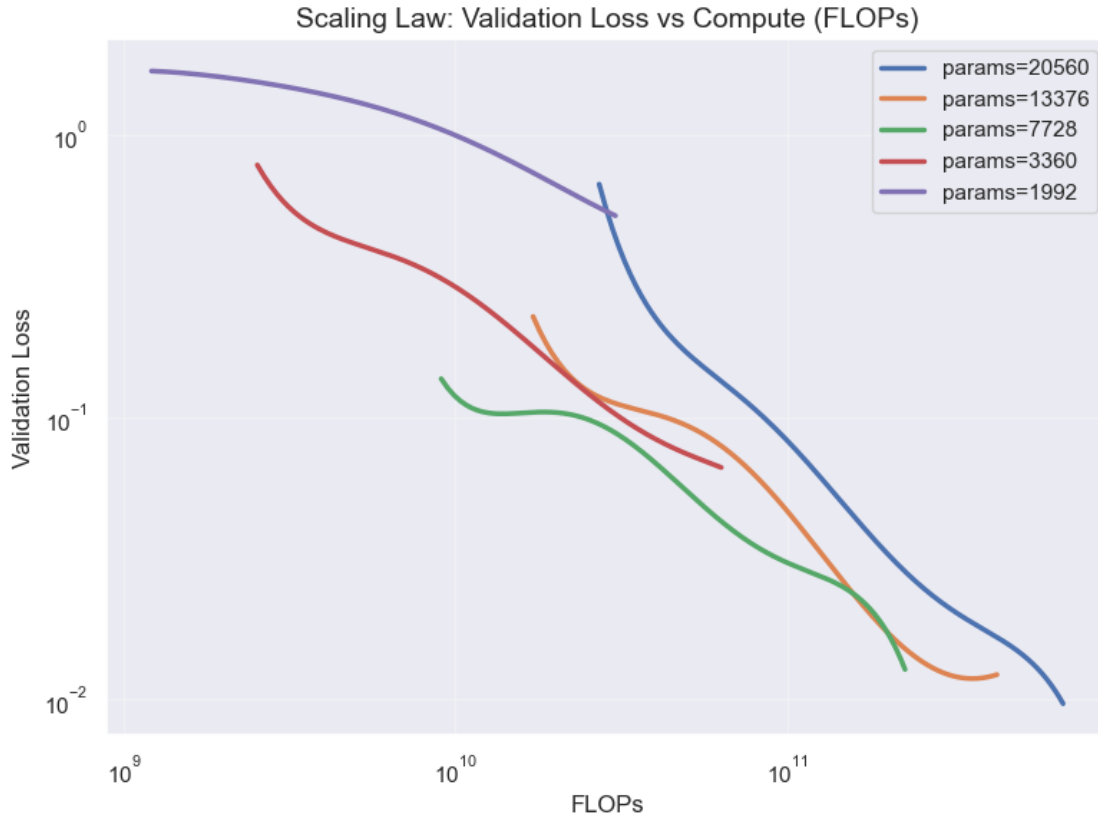
Epoch 1		Train loss: 1.785961		Val loss 1.685956
---------	--	----------------------	--	-------------------

Epoch 2	Train loss: 1.617972	Val loss 1.548851
Epoch 3	Train loss: 1.492224	Val loss 1.437632
Epoch 4	Train loss: 1.381100	Val loss 1.317770
Epoch 5	Train loss: 1.287916	Val loss 1.231228
Epoch 6	Train loss: 1.197418	Val loss 1.149385
Epoch 7	Train loss: 1.117825	Val loss 1.080033
Epoch 8	Train loss: 1.048070	Val loss 1.004309
Epoch 9	Train loss: 0.981025	Val loss 0.961329
Epoch 10	Train loss: 0.922533	Val loss 0.883386
Epoch 11	Train loss: 0.867679	Val loss 0.886065
Epoch 12	Train loss: 0.821292	Val loss 0.824073
Epoch 13	Train loss: 0.779774	Val loss 0.752169
Epoch 14	Train loss: 0.742627	Val loss 0.732572
Epoch 15	Train loss: 0.712286	Val loss 0.699841
Epoch 16	Train loss: 0.682646	Val loss 0.668081
Epoch 17	Train loss: 0.657836	Val loss 0.670059
Epoch 18	Train loss: 0.635605	Val loss 0.646111
Epoch 19	Train loss: 0.615669	Val loss 0.601949
Epoch 20	Train loss: 0.592914	Val loss 0.589267
Epoch 21	Train loss: 0.578662	Val loss 0.573499
Epoch 22	Train loss: 0.561698	Val loss 0.557479
Epoch 23	Train loss: 0.547712	Val loss 0.537994
Epoch 24	Train loss: 0.532104	Val loss 0.532714
Epoch 25	Train loss: 0.515401	Val loss 0.517307

```
[39]: plot_scaling_law(all_results)

      plot_scaling_law_poly(all_results)
```





3.2 Step 2: Optimal Number of Parameters

Using the previously created training runs, we'll fit quadratic functions to the (parameter count, validation loss) pairs to estimate the optimal number of parameters for a given compute budget. We'll then fit a line to this set of optimal parameters to estimate the optimal number of parameters for a compute budget of $1e15$.

```
[40]: def find_optim_params(params: torch.Tensor, loss: torch.Tensor):
    x = np.array(params.cpu())
    y = np.array(loss.cpu())

    p = np.polyfit(np.log10(x), y, 2)
    optimal_log_params = -(p[1]) / (2 * p[0])
    optimal_params = 10 ** (optimal_log_params)

    return optimal_params

target_flops = [8e9, 16e9, 32e9, 64e9, 128e9]
colors = ["#E41A1C", "#377EB8", "#4DAF4A", "#984EA3", "#FF7F00"]
params_vs_loss = []
```

```

optimal_params = []

for target in target_flops:
    parameters = []
    val_loss = []

    for params, results in all_results.items():
        loss = interpolate(results["val_loss"], results["flops"], target, deg=4)

        # interpolate returns None if target is out of range
        if loss != None:
            parameters.append(params)
            val_loss.append(loss)

    sorted_indices = torch.argsort(torch.tensor(parameters)).to(device)
    parameters = torch.tensor(parameters).to(device)[sorted_indices]
    val_loss = torch.tensor(val_loss).to(device)[sorted_indices]
    try:
        optimal_params.append(find_optim_params(parameters, val_loss))
        params_vs_loss.append((parameters.tolist(), val_loss.tolist()))
    except Exception as e:
        print(
            "\n\n{optimal_params=}\n\n{params_vs_loss=}\n\n{parameters=}\n\n{val_loss=}\n\n{target=}"
        )

```

```

/var/folders/dj/k5kk5wk90vvcnb_wphpw490r0000gn/T/ipykernel_92315/2694499551.py:2
: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so
passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword
arguments. To learn more, see the migration guide
https://numpy.org/devdocs/numpy\_2\_0\_migration\_guide.html#adapting-to-changes-in-the-copy-keyword

```

```

x = np.array(params.cpu())
/var/folders/dj/k5kk5wk90vvcnb_wphpw490r0000gn/T/ipykernel_92315/2694499551.py:3
: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so
passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword
arguments. To learn more, see the migration guide
https://numpy.org/devdocs/numpy\_2\_0\_migration\_guide.html#adapting-to-changes-in-the-copy-keyword
y = np.array(loss.cpu())

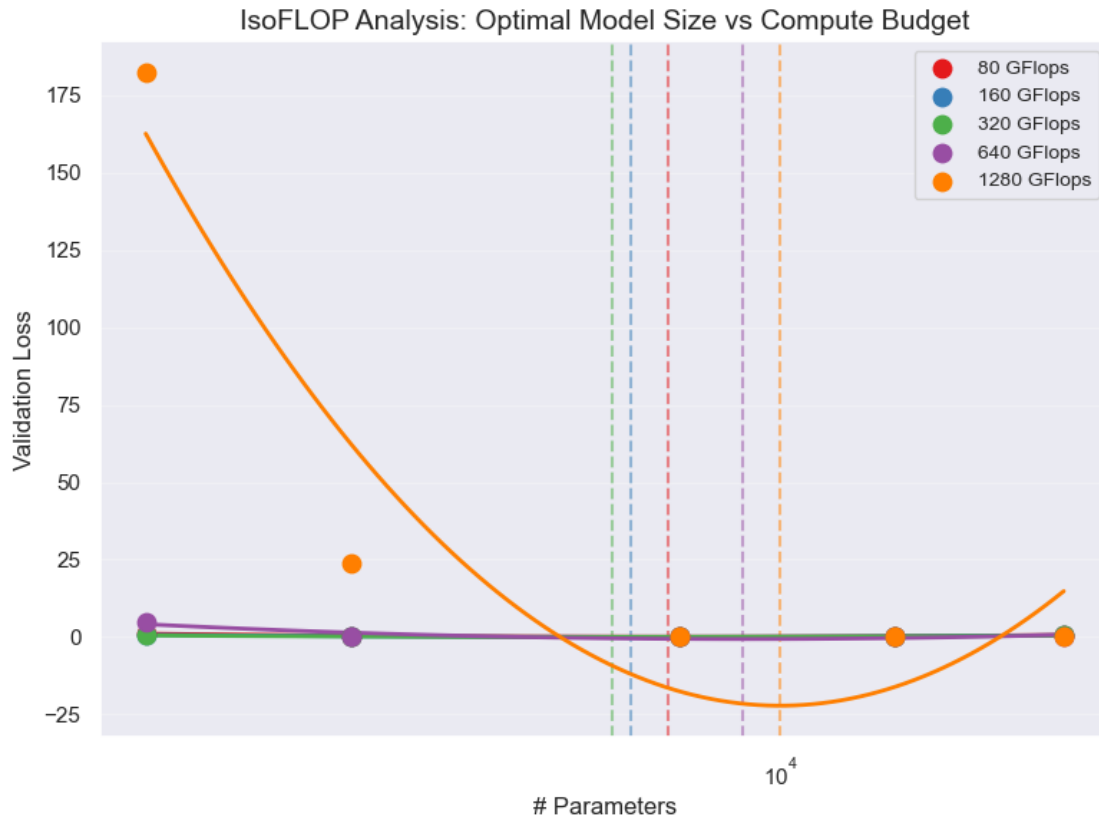
```

```
[41]: plot_isoflop(target_flops, colors, params_vs_loss, optimal_params)
```

```

8.00 GFlops: Optimal params = 7,514
16.00 GFlops: Optimal params = 6,831
32.00 GFlops: Optimal params = 6,524
64.00 GFlops: Optimal params = 9,090
128.00 GFlops: Optimal params = 9,992

```



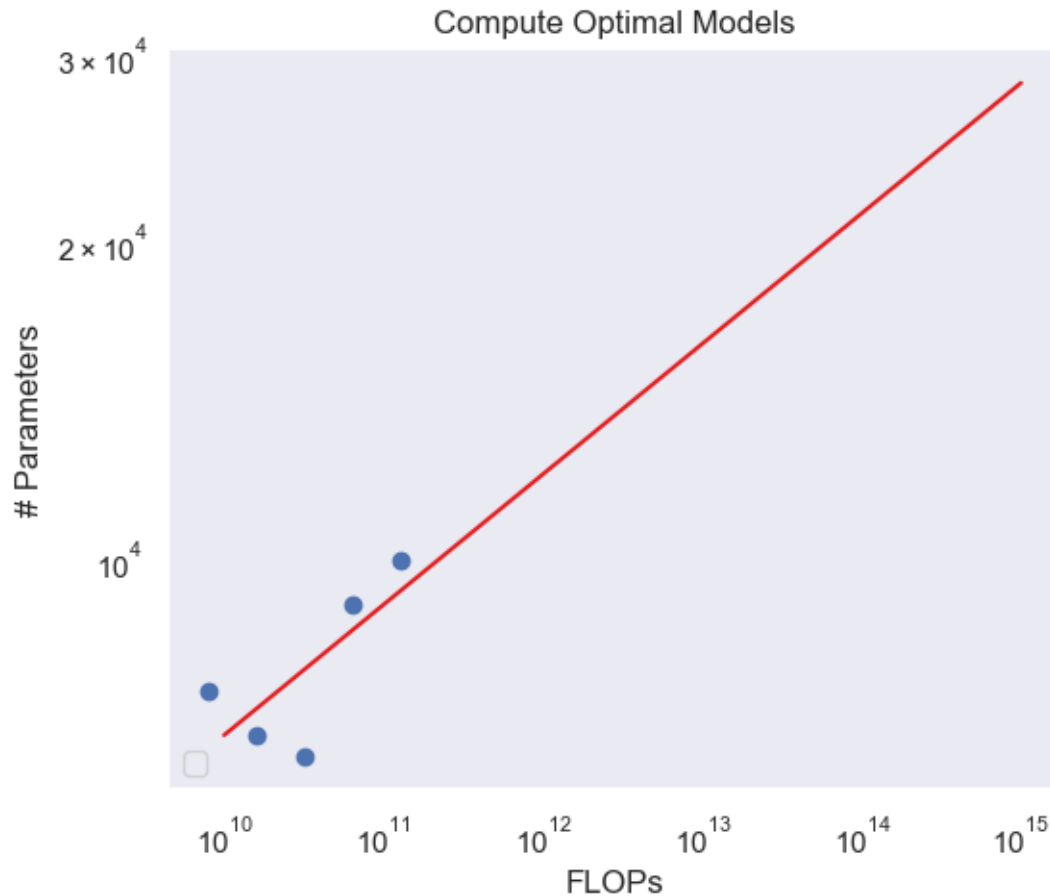
```
[42]: plot_flops_params(target_flops, optimal_params)
```

```
y = 0.12346471733672994x + 2.59959596291159
```

```
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:282:
```

```
UserWarning: No artists with labels found to put in legend. Note that artists
whose label start with an underscore are ignored when legend() is called with no
argument.
```

```
ax.legend(loc="lower left")
```



3.3 Step 3: Optimal Number of Tokens

Finally, we'll perform a similar analysis to the previous step, only this time determining the optimal number of tokens for a budget of $1e15$ flops.

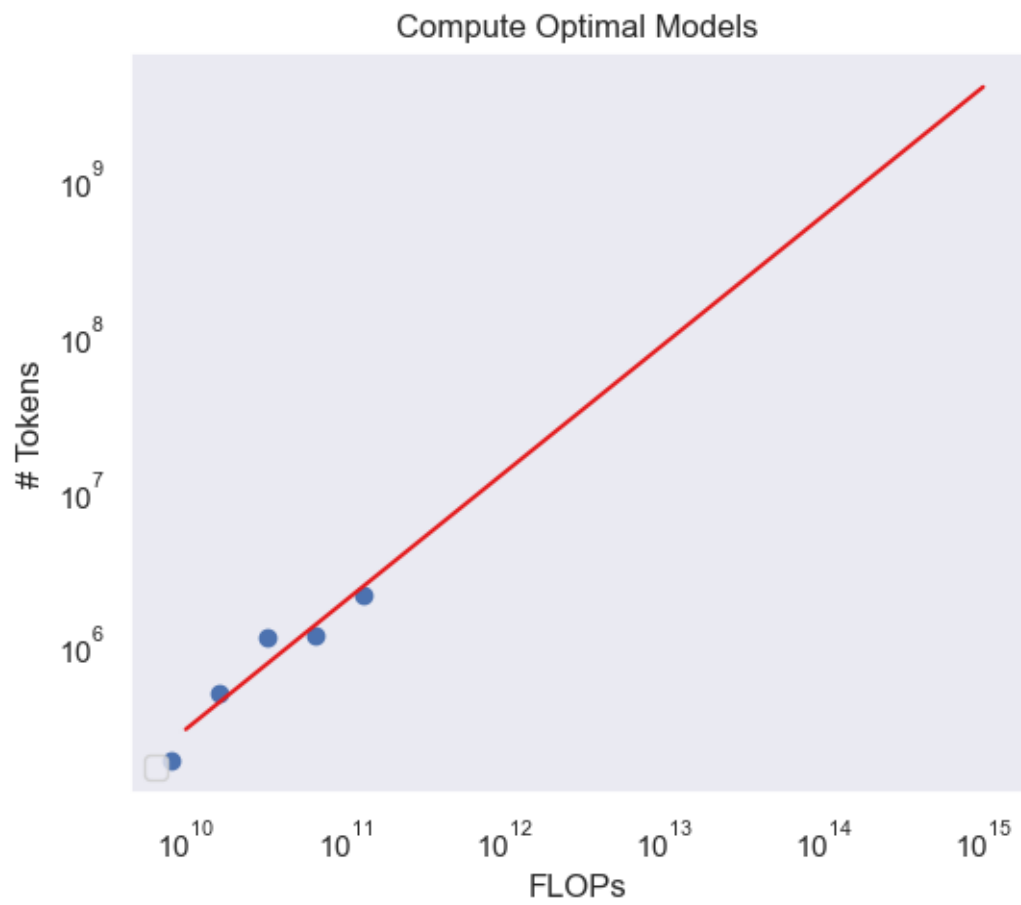
```
[43]: plot_flops_tokens(target_flops, optimal_params, all_results)
```

```
y = 0.8263812499134265x + -2.804966406377365
```

```
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:323:
```

```
UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.
```

```
ax.legend(loc="lower left")
```



[]: